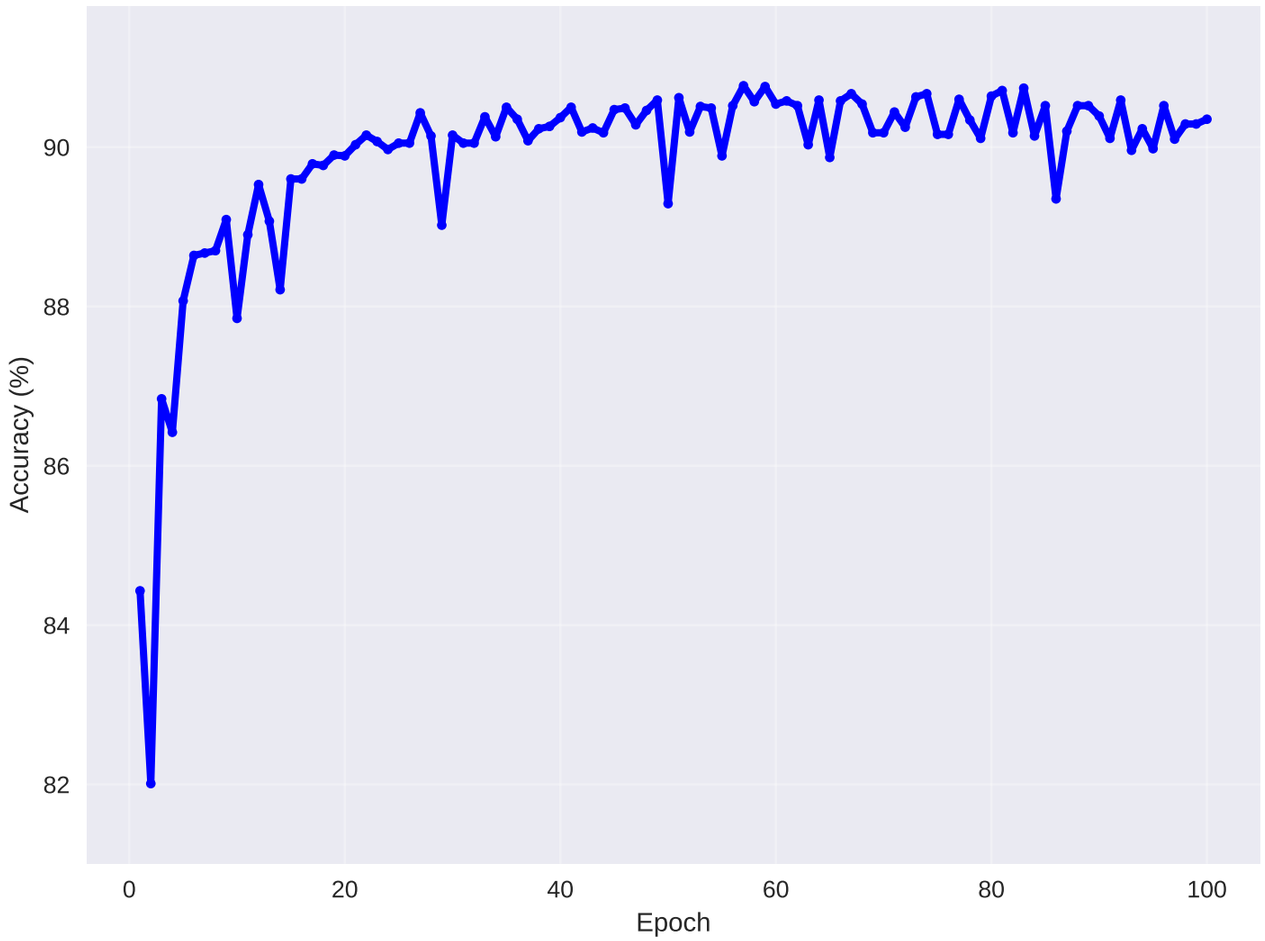
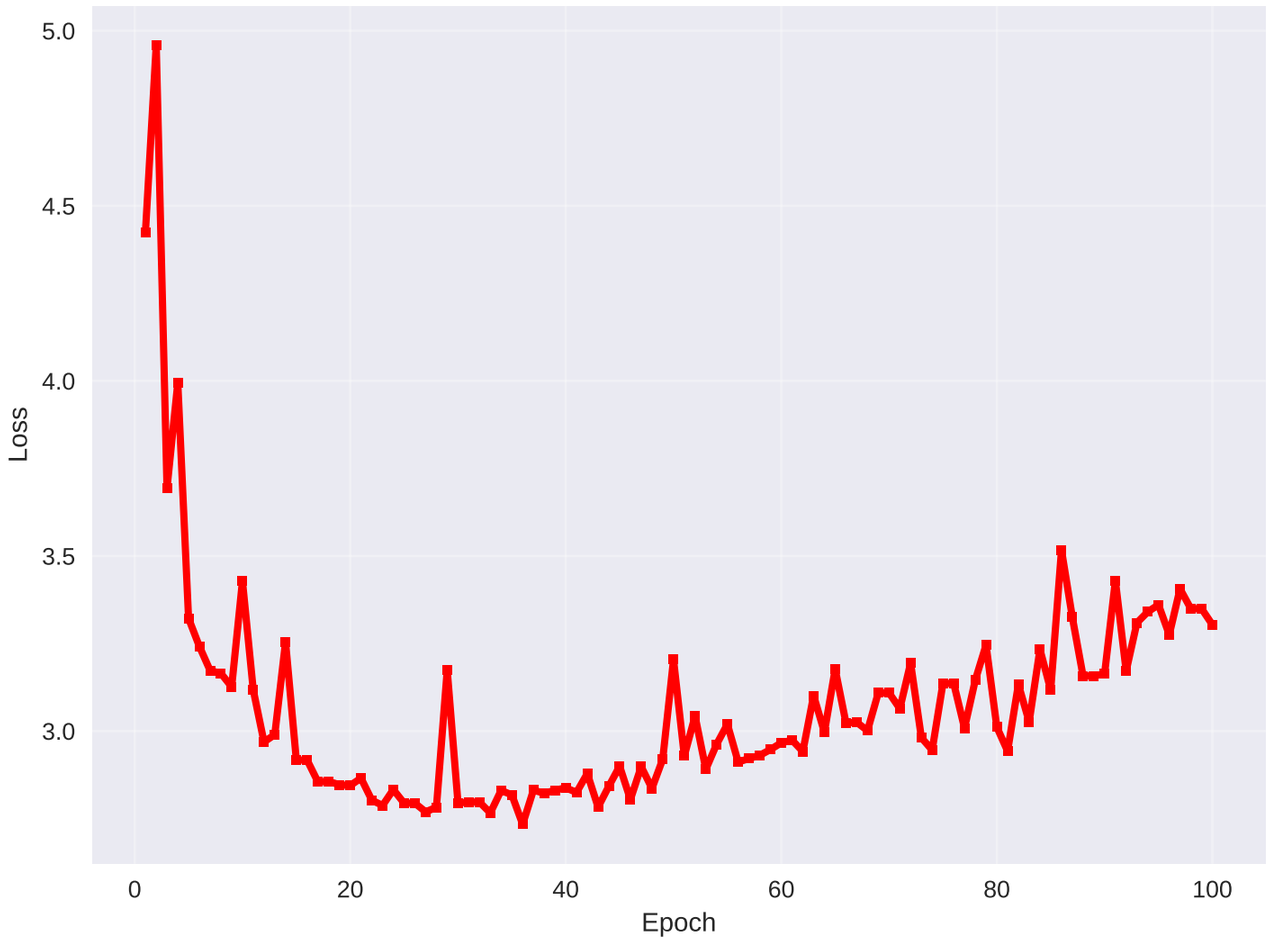
# Federated Learning - Main Metrics Overview

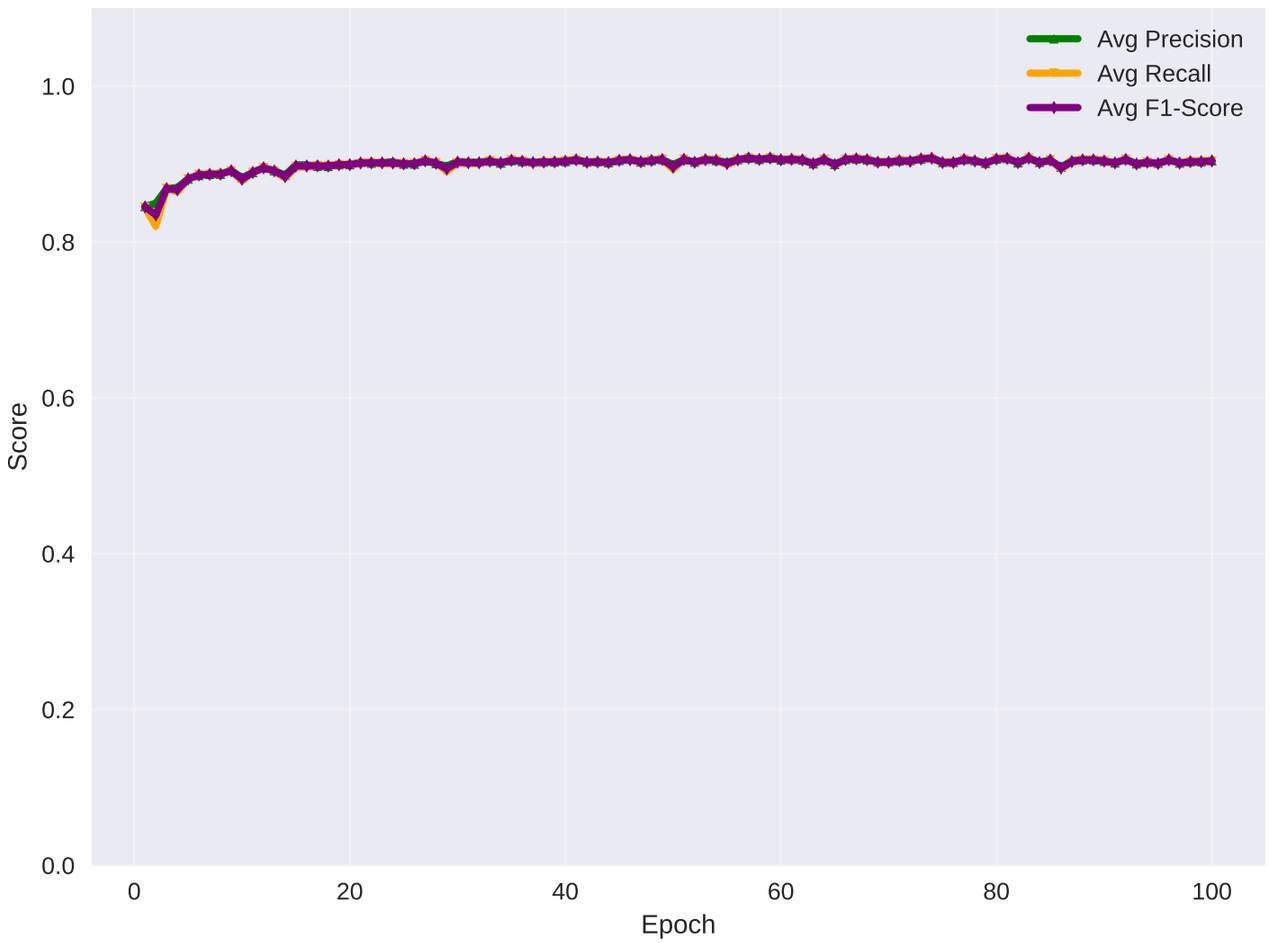
□ Global Model Accuracy Evolution



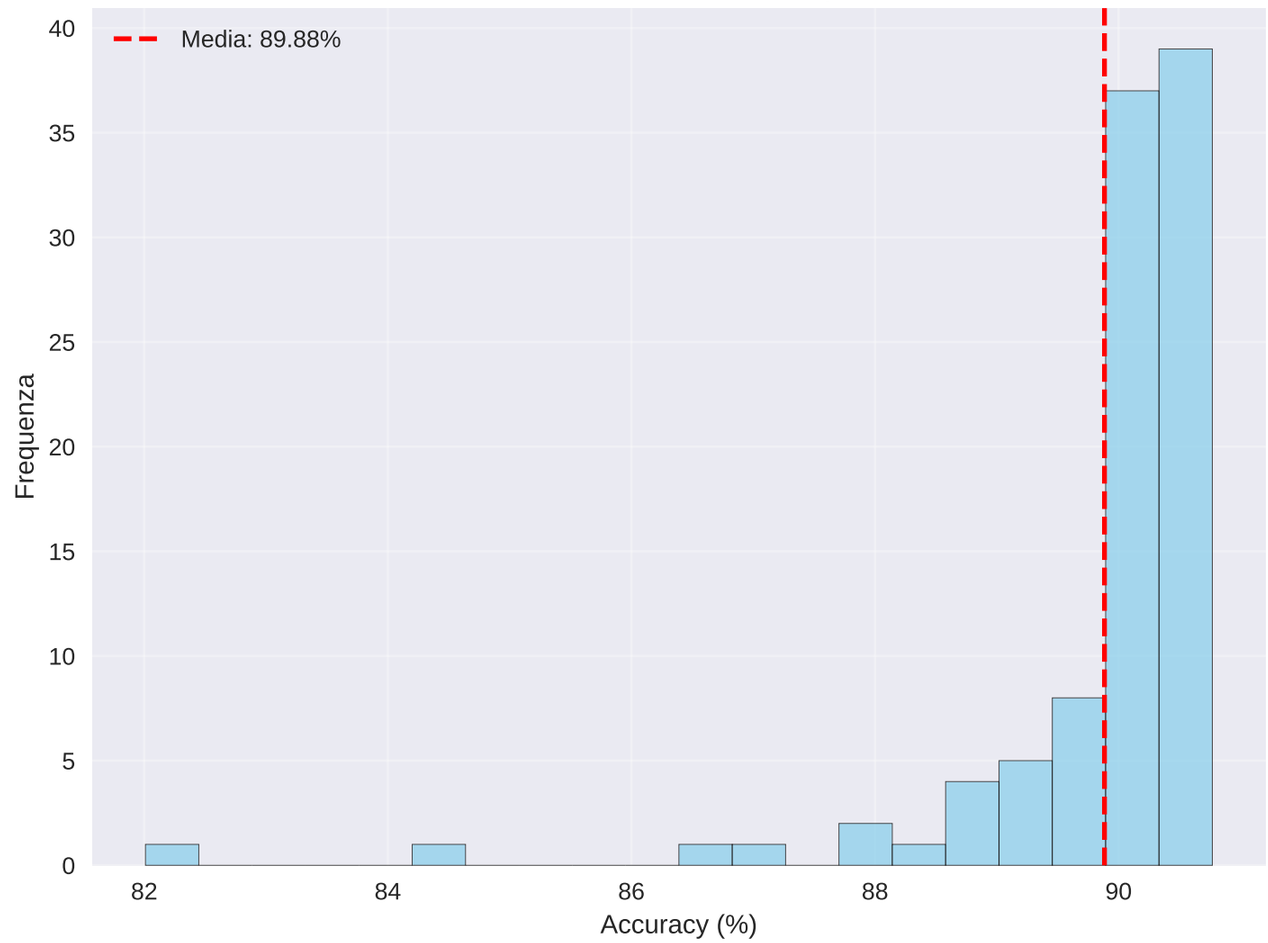

□ Global Model Loss Evolution




□ Average Precision, Recall & F1-Score

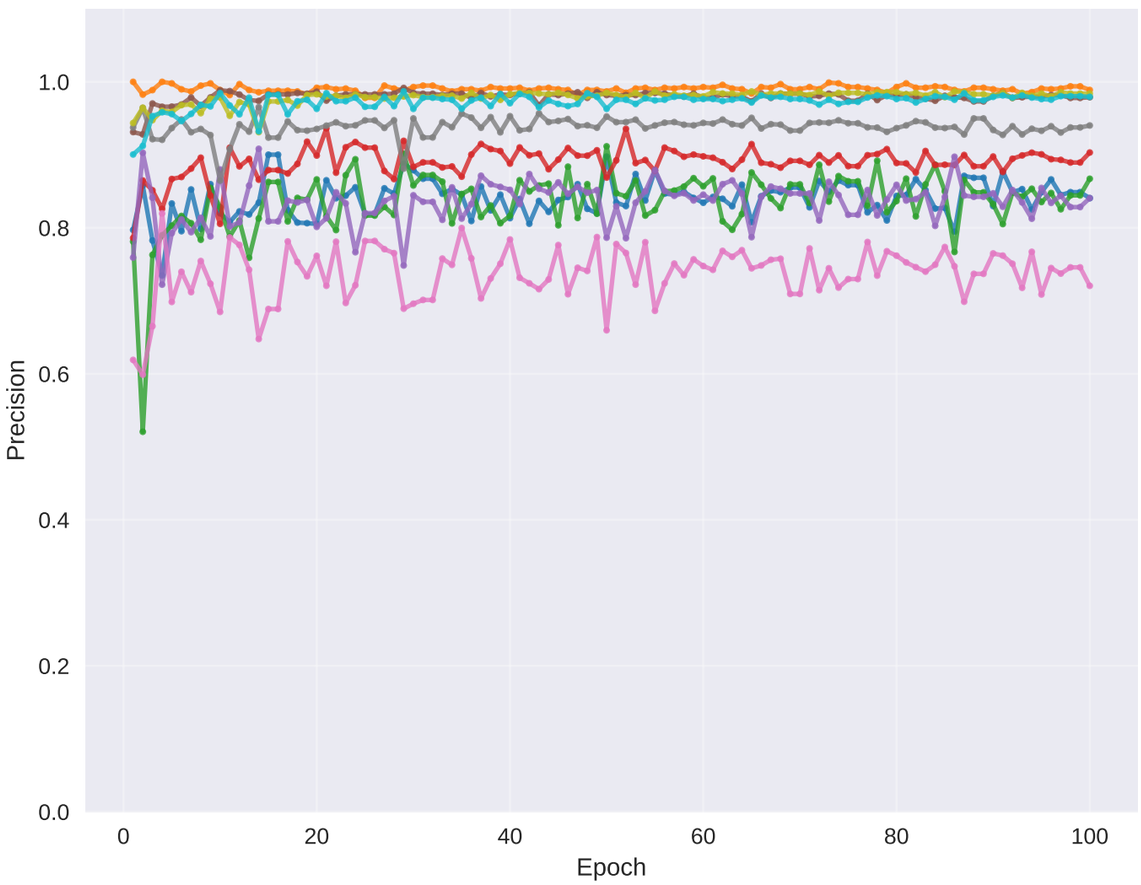



□ Distribuzione Accuracy

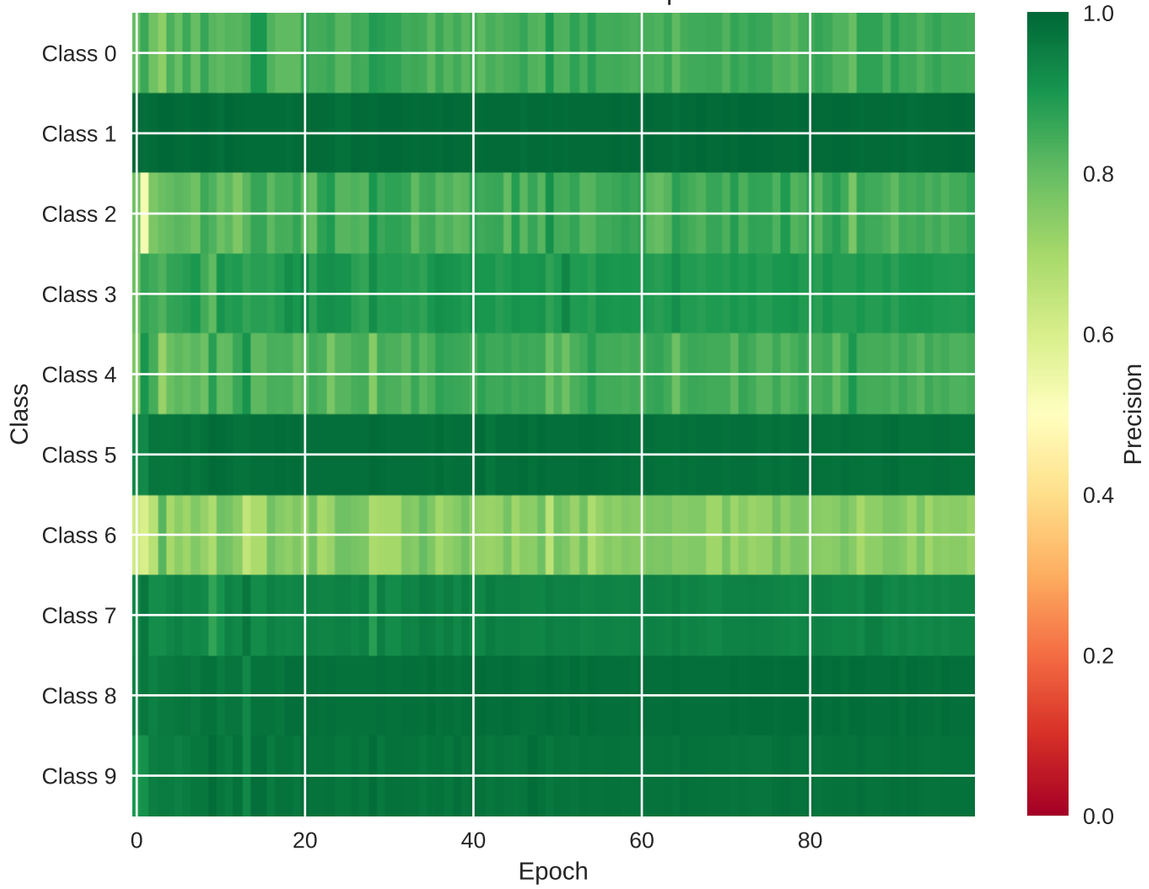


# Per-Class Precision Analysis

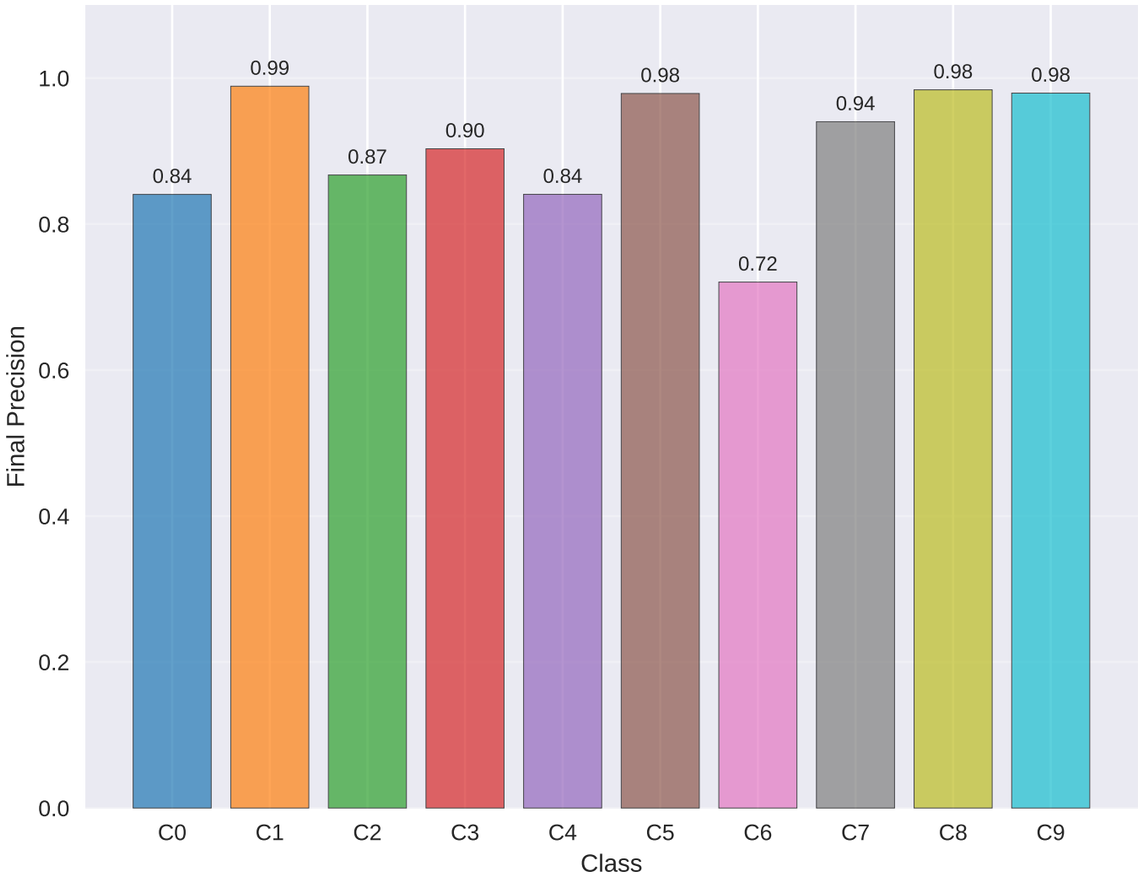
□ Per-Class Precision Evolution



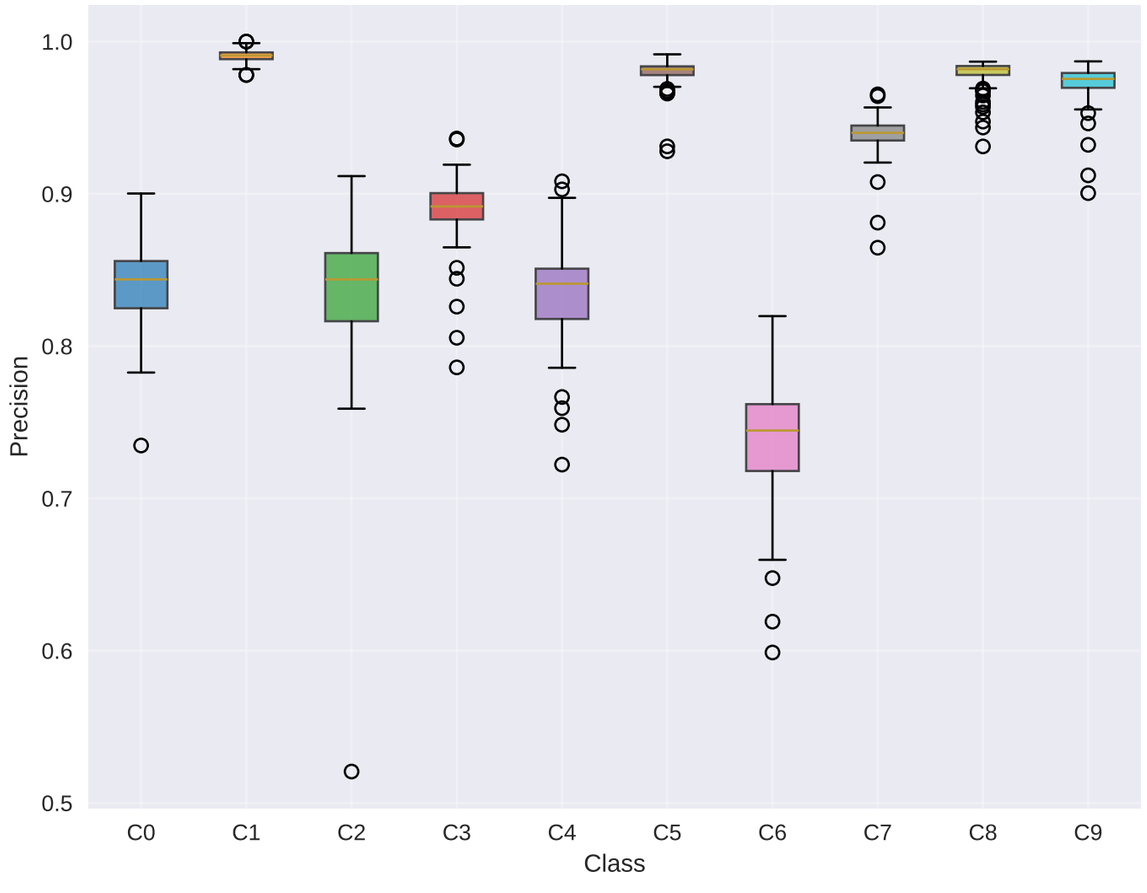

□ Precision Heatmap




□ Final Precision per Class

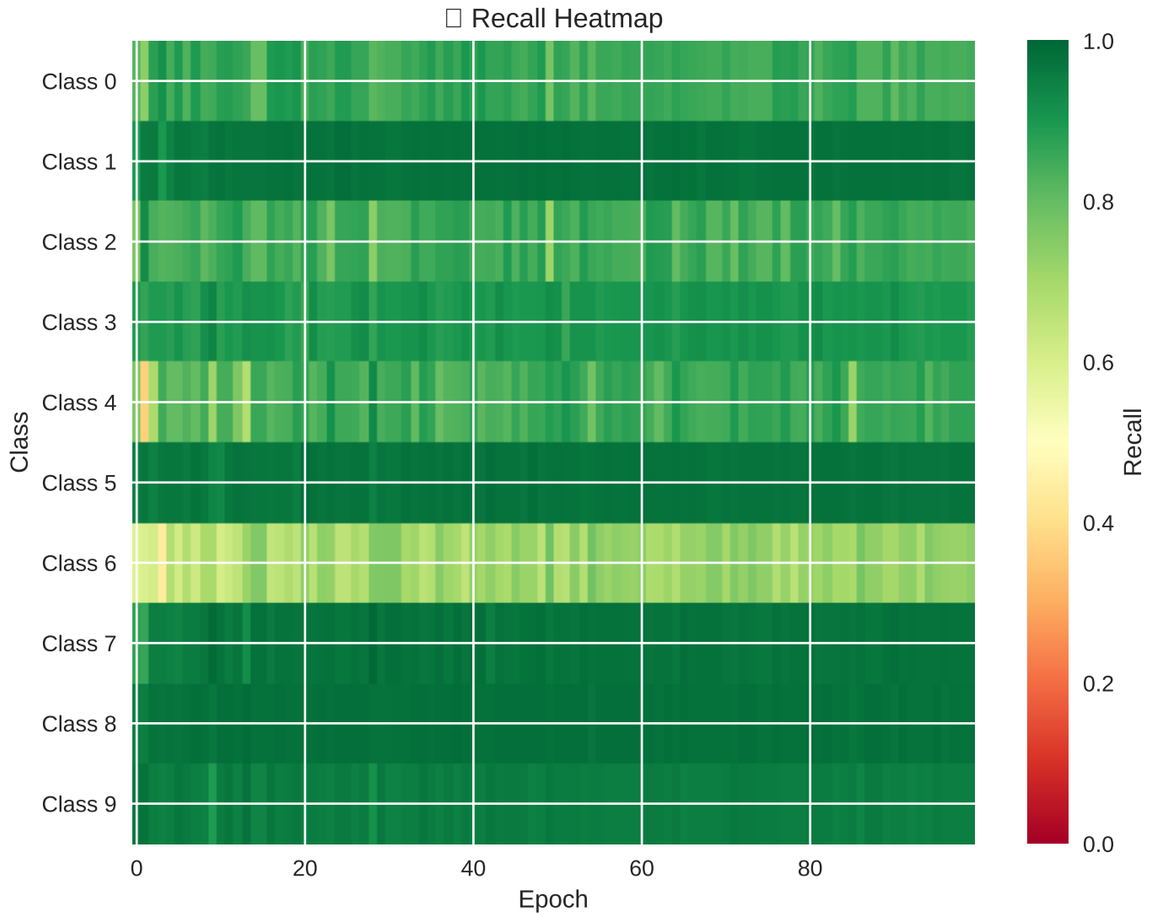


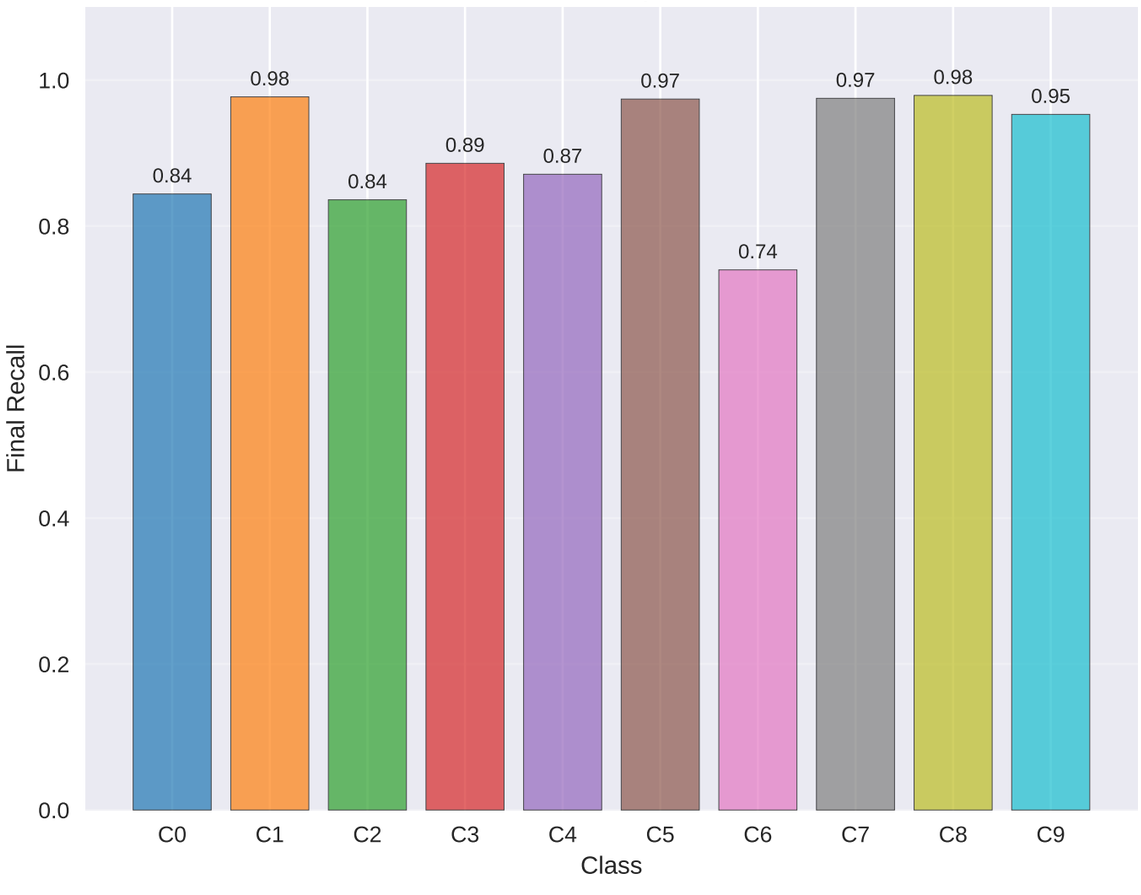

□ Precision Distribution per Class

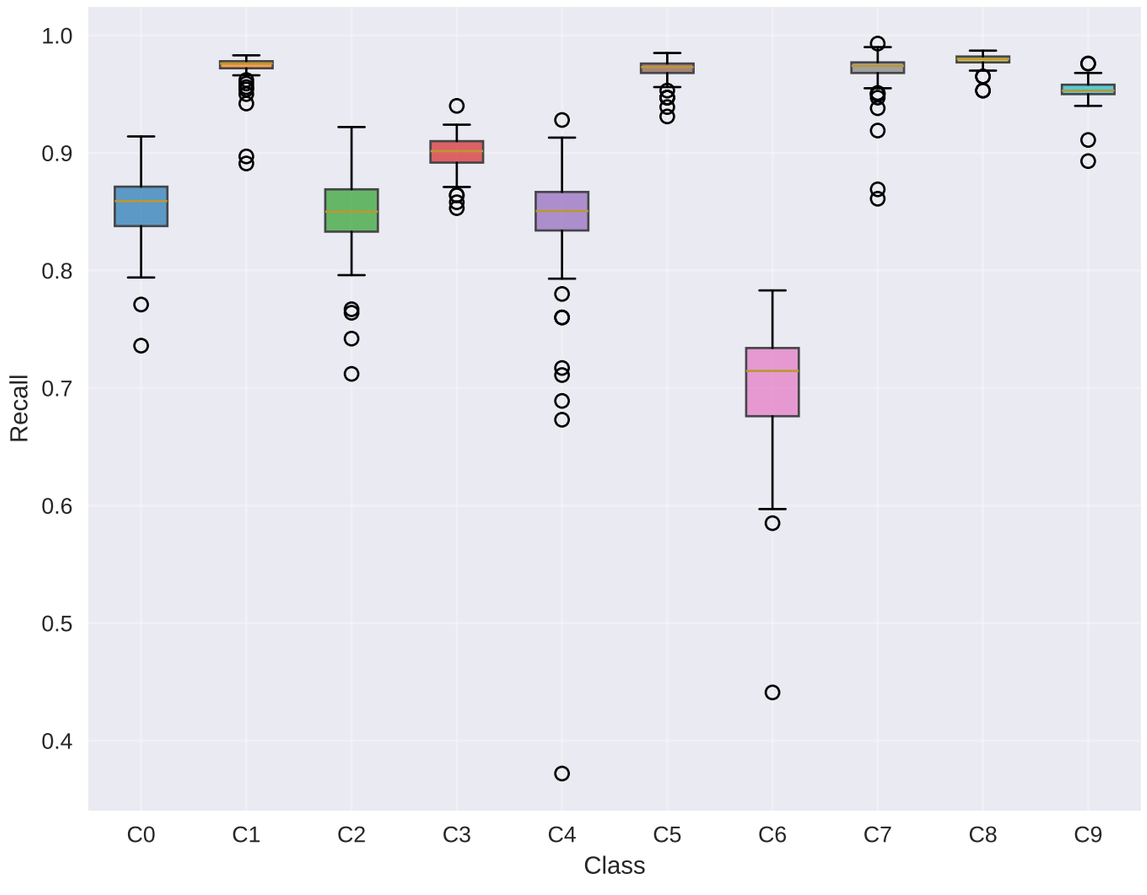


# Per-Class Recall Analysis

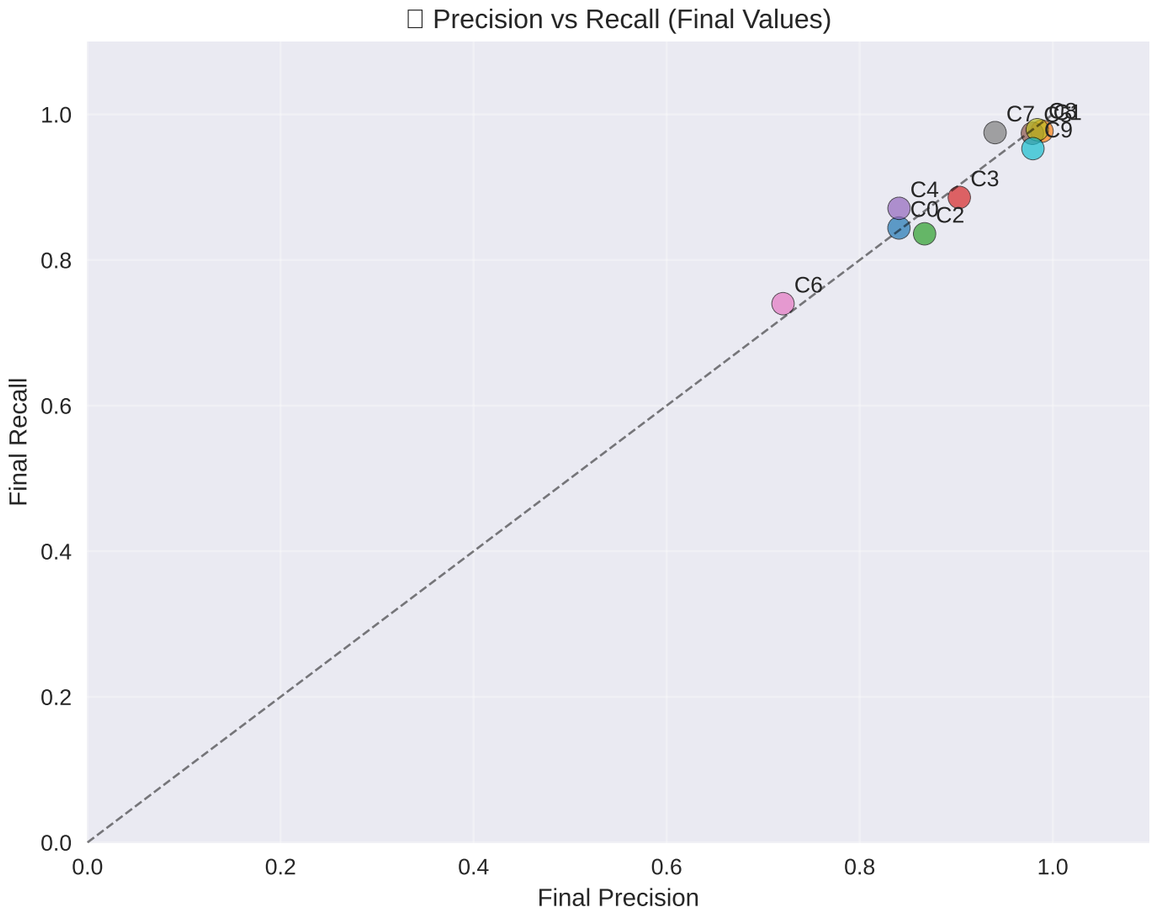


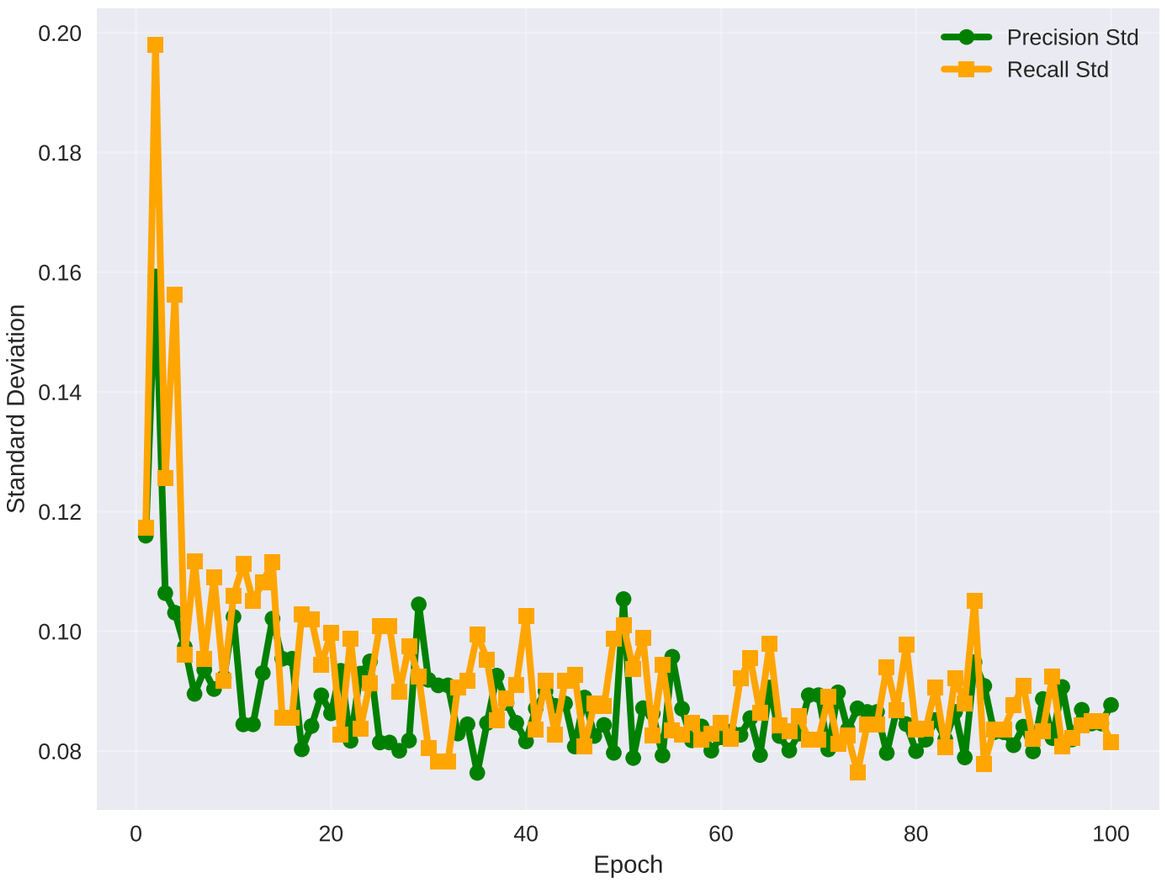



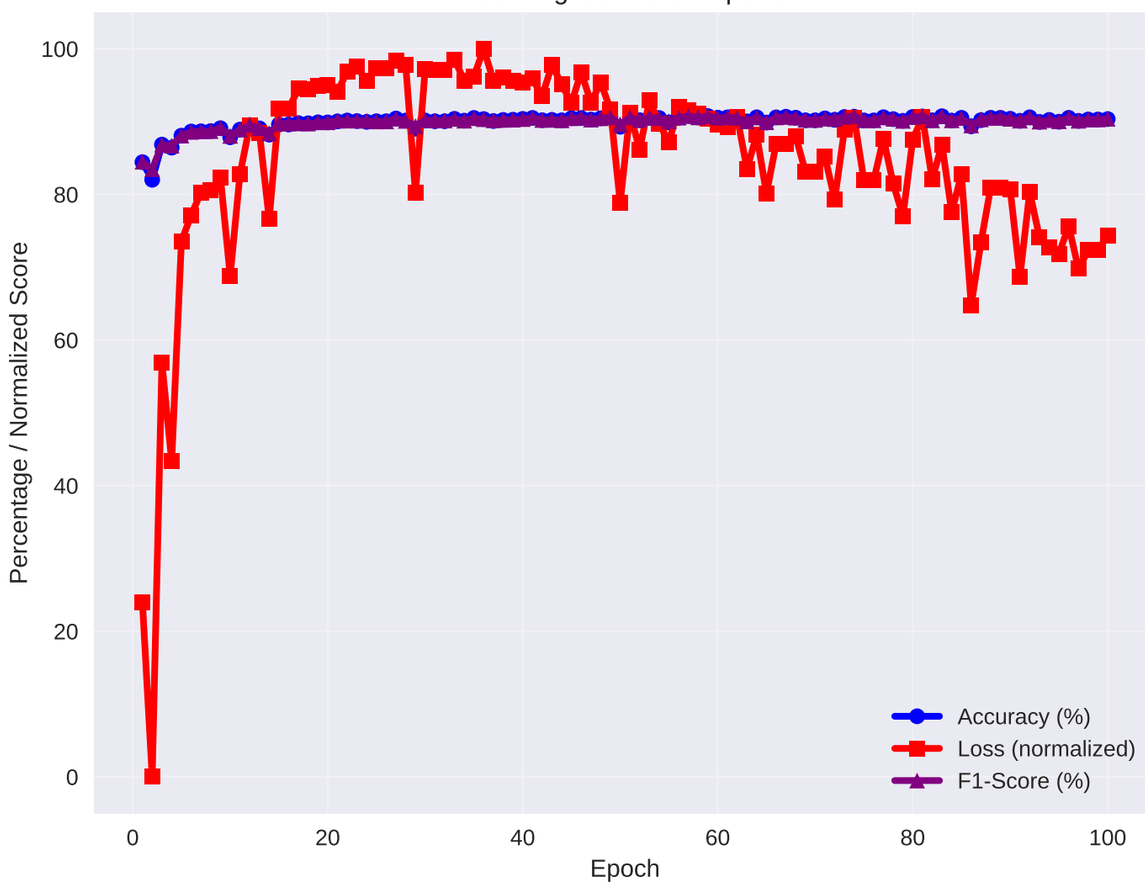
# Comparative Analysis









# Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-29 12:53:56
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.35%
• Loss: 3.3037
• Average Precision: 0.904
• Average Recall: 0.903
• Average F1-Score: 0.904

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +5.92%
• Riduzione Loss: +1.1215
• Stabilità Accuracy (CV): 0.0138

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.989)
• Peggiore Precision: Classe 6 (0.721)
• Migliore Recall: Classe 8 (0.979)
• Peggiore Recall: Classe 6 (0.740)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.841, R=0.844, F1=0.842
• Classe 1: P=0.989, R=0.977, F1=0.983
• Classe 2: P=0.867, R=0.836, F1=0.851
• Classe 3: P=0.903, R=0.886, F1=0.894
• Classe 4: P=0.841, R=0.871, F1=0.856
• Classe 5: P=0.979, R=0.974, F1=0.976
• Classe 6: P=0.721, R=0.740, F1=0.730
• Classe 7: P=0.940, R=0.975, F1=0.957
• Classe 8: P=0.984, R=0.979, F1=0.981
• Classe 9: P=0.979, R=0.953, F1=0.966

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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